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Case 39-2023: A 43-Year-Old Woman with Chronic Diarrhea, Hair Loss, and Nail and Skin Changes

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PRESENTATION OF CASE

Dr. W.K. Claudia Wu: A 43-year-old woman was admitted to the Prince of Wales Hospital of the Chinese University of Hong Kong because of abdominal pain and diarrhea.

The patient had been well until 6 months before this admission, when colicky, diffuse abdominal pain developed. She also had nonbloody diarrhea that occurred up to 10 times daily. During the subsequent 6 months, the patient noticed decreased appetite and weight loss of 5 kg. The symptoms did not abate, and her family encouraged her to seek medical evaluation.

On evaluation, the patient reported gradually darkening skin on the face, arms, and legs, as well as hair thinning and changes in the appearance of all the fingernails and toenails during the previous 4 months. There was no nausea, vomiting, hematemesis, hematochezia, fevers, or night sweats.

The patient had a history of polycystic ovarian syndrome. Ten months before this admission, she had sought evaluation by a traditional Chinese medicine practitioner because of irregular menses. She was given a “homemade” traditional Chinese medicine that contained unknown ingredients. The patient took the remedy for 4 months and stopped taking it when the abdominal pain and diarrhea developed 6 months before this admission. She did not take other medications and had no known drug allergies. The patient lived with her husband and son in an urban area of Hong Kong, and she did not work outside the home. She did not vape or smoke cigarettes, and she did not drink alcohol or use illicit drugs. Her mother and father were healthy; there was no family history of gastroenterologic, autoimmune, or hereditary diseases.

On evaluation, the temporal temperature was 37.2°C, the blood pressure 120/65 mm Hg, and the heart rate 85 beats per minute. The body-mass index (the weight in kilograms divided by the square of the height in meters) was 20. Examinations

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Figure 1. Photographs of the Scalp and Skin.

Nonscarring alopecia is observed over the vertex of the scalp (Panel A); a hair-pull test was negative. Symmetric brown macules are present over the lower cheeks and malar regions of the face (Panels B and C), and diffuse brown-gray hyperpigmented macules are present over the palms and the palmar aspects of the fingers (Panels D and E). Onychodystrophy and onycholysis are present on all the fingernails (Panels E and F) and toenails.

of the thyroid, heart, and lungs were normal. The abdomen was soft and nontender. There was no cervical, axillary, or inguinal lymphadenopathy. Moderate pitting edema was observed in the lower legs, up to the mid-calf.

Dr. W.S. Agnes Chan: On dermatologic examination, the patient had patchy hair loss, with a large patch of alopecia at the vertex of the scalp; there was no scarring of the scalp (Fig. 1A). A hair-pull test (in which 40 to 60 strands of closely grouped hair are gently pulled, with more than 10% of hairs pulled out indicating a positive result) was negative. There were symmetric brown macules over the lateral lower cheeks and malar regions of the face (Fig. 1B and 1C), and hyperpigmentation and diffuse brown-gray macules were observed over the palms and the palmar aspects of the fingers (Fig. 1D and 1E). Onychodystrophy was present on all the fingernails (Fig. 1E) and toenails. There was onycholy-

sis with distal separation of the nail plates from the nail beds, thinned nail plates, distal nail-plate crumbling, and onychomadesis with proximal separation of the nail plates from the nail matrices (Fig. 1E and 1F).

Dr. Wu: The blood level of hemoglobin was 10.0 g per deciliter (reference range, 11.9 to 15.1) with a low mean corpuscular volume. The white-cell count, C-reactive protein level, and erythrocyte sedimentation rate in blood were normal. The blood level of potassium was 2.9 mmol per liter (reference range, 3.5 to 5.0), ionized calcium 0.85 mmol per liter (reference range, 1.13 to 1.32), phosphate 0.52 mmol per liter (reference range, 0.72 to 1.43), and zinc 8.8 μ mol per liter (reference range, 10.7 to 18.0). The blood level of albumin was 14 g per liter (reference range, 33 to 48). The protein level in a spot urine sample was normal. A stool culture showed normal enteric flora. Testing of a stool specimen for *Clostridium*

difficile antigen and Shiga toxin was negative, as was a stool examination for ova and parasites. Other laboratory test results are shown in Table 1.

The patient was admitted to the hospital, and a diagnostic test was performed.

DIFFERENTIAL DIAGNOSIS

Dr. Jimmy C.T. Lai: I was involved in the care of this patient and am aware of the final diagnosis. This 43-year-old woman presented with weight loss, chronic diarrhea, and multiple cutaneous abnormalities, including changes to the nails, hair loss, and skin hyperpigmentation that developed after the use of a traditional Chinese medicine containing unknown ingredients. There was laboratory evidence of malabsorption and electrolyte disturbance, findings consistent with protein-losing enteropathy. In constructing a differential diagnosis, I will consider diseases that cause protein-losing enteropathy and are associated with dermatologic manifestations.

AUTOIMMUNE DISEASES

Inflammatory bowel disease, systemic vasculitis, systemic lupus erythematosus (SLE), and celiac disease are autoimmune disorders that may cause chronic gastrointestinal symptoms. Inflammatory bowel disease, especially Crohn's disease, manifests with abdominal pain and chronic diarrhea that can be nonbloody. If there is small-bowel involvement, malabsorption can occur, with resultant electrolyte abnormalities, micronutrient deficiencies, and weight loss, all of which were seen in this patient. Extraintestinal manifestations of Crohn's disease that affect the skin include erythema nodosum (characterized by raised, erythematous, tender nodules appearing mainly on the shins), pyoderma gangrenosum (characterized by ulcerations with violaceous edges appearing mainly on the legs), and Sweet's syndrome (characterized by a sudden onset of painful nodules and fevers).¹ This patient's skin changes are not consistent with these features of Crohn's disease.

A systemic vasculitis, such as Behçet's disease, or intestinal vasculitis can also lead to chronic diarrhea, protein-losing enteropathy, and rashes. However, the absence of a purpuric rash and of oral, ophthalmic, or genital ulcers in this patient makes this diagnosis unlikely.²

SLE has been associated with protein-losing

Table 1. Laboratory Data.*

Variable	Reference Range†	On Admission
White-cell count (per μ l)	4000–9700	9000
Hemoglobin (g/dl)	11.9–15.1	10.0
Mean corpuscular volume (fl)	83–98	74
Platelet count (per μ l)	150,000–384,000	454,000
Erythrocyte sedimentation rate (mm/hr)	5–34	19
C-reactive protein (mg/liter)	0.0–9.9	3.7
Sodium (mmol/liter)	137–144	141
Potassium (mmol/liter)	3.5–5.0	2.9
Urea nitrogen (mmol/liter)	2.6–6.6	2.4
Creatinine (mg/dl)	0.55–0.94	0.45
Magnesium (mmol/liter)	0.66–1.07	0.89
Phosphate (mmol/liter)	0.72–1.43	0.52
Calcium (mg/dl)	2.15–2.55	1.30
Ionized calcium (mmol/liter)	1.13–1.32	0.85
Albumin (g/liter)	33–48	14
Globulin (g/liter)	20–35	29
Total protein (g/liter)	65–82	43
Alanine aminotransferase (U/liter)	<47	10
Aspartate aminotransferase (U/liter)	<35	13
Alkaline phosphatase (U/liter)	33–84	56
Total bilirubin (μ mol/liter)	0–17	3
Zinc (μ mol/liter)	10.7–18.0	8.8
Iron (μ mol/liter)	6–35	3
Total iron-binding capacity (μ g/dl)	229–430	212
Transferrin saturation (%)	16–45	8
Ferritin (μ g/liter)	13–150	5
Vitamin B ₁₂ (pg/ml)	>24	42
Folate (nmol/liter)	7.0–36.4	11.8

* To convert the values for creatinine to micromoles per liter, multiply by 88.4. To convert the values for calcium to millimoles per liter, multiply by 0.250. To convert the values for total iron-binding capacity to micromoles per liter, multiply by 0.1791. To convert the values for vitamin B₁₂ to picomoles per liter, multiply by 0.7378.

† Reference values are affected by many variables, including the patient population and the laboratory methods used. The ranges used at Prince of Wales Hospital are for adults who are not pregnant and do not have medical conditions that could affect the results. They may therefore not be appropriate for all patients.

enteropathy.^{3,4} However, with the exception of nonscarring alopecia, this patient does not meet clinical criteria that would indicate a diagnosis of SLE.⁵ Nonetheless, additional immunologic tests should be performed to rule out SLE.

Celiac disease can be associated with derma-

titis herpetiformis, a rash characterized by itchy papules and vesicles that are distributed symmetrically over the shoulders, elbows, and knees. Because celiac disease is rare in the Asian population⁶ and this patient's skin findings would be atypical of celiac disease, this diagnosis is also unlikely.

PRIMARY ADRENAL INSUFFICIENCY

Primary adrenal insufficiency (also known as Addison's disease) is associated with a variety of clinical manifestations, ranging from malaise and weight loss to nonspecific gastrointestinal symptoms, such as abdominal pain and diarrhea, and can cause amenorrhea in female patients. Cortisol insufficiency causes overproduction of proopiomelanocortin and subsequent excessive melanocyte-stimulating hormone, leading to some degree of skin hyperpigmentation in most patients. This patient has weight loss, abdominal pain, diarrhea, menstrual abnormalities, and skin hyperpigmentation; however, Addison's disease is not associated with hair loss or nail changes.

HEAVY-METAL POISONING

The patient's symptoms began soon after the use of a traditional Chinese medicine with unknown ingredients, which suggested the possibility of heavy-metal poisoning caused by ingestion of either arsenic or lead.⁷⁻⁹ Arsenic poisoning can cause skin hyperpigmentation but does not typically result in hair and nail changes. Lead poisoning can cause gastrointestinal symptoms such as abdominal pain but not skin changes. Chronic arsenic or lead poisoning may cause other systemic symptoms, such as sensorimotor polyneuropathy, but those symptoms are not present in this patient.

CANCER AND AMYLOIDOSIS

Cancer is always an important consideration in the differential diagnosis of unintentional weight loss. Colorectal cancer is possible in this patient, despite her relatively young age and the absence of risk factors. In particular, rectal cancer, the incidence of which is rising among younger adults,^{10,11} can cause diarrhea. Gastrointestinal lymphoma can lead to chronic diarrhea and protein-losing enteropathy, along with cutaneous involvement, although such manifestations are uncommon.

Amyloidosis can affect many organs and has a variety of manifestations. Systemic amyloidosis can cause rash, whereas amyloidosis that affects the gastrointestinal tract can lead to protein-losing enteropathy, malabsorption, and constitutional symptoms.¹² Amyloidosis is a rare disease, and a biopsy of the affected sites would be needed to confirm the presence of amyloid deposition.

OTHER DISEASES

Despite the absence of fever in this patient, infection is a consideration. Tuberculosis is prevalent in the area where this patient lives, and gastrointestinal tuberculosis can cause chronic diarrhea and constitutional symptoms such as weight loss. However, the absence of fevers and night sweats, as well as the presence of hair and nail changes, are not consistent with the diagnosis of tuberculosis.

The Cronkhite–Canada syndrome (CCS), a rare, apparently noninherited hamartomatous polyposis syndrome, may account for many of the features of this patient's clinical presentation. This syndrome was first reported by Dr. L.W. Cronkhite, Jr., and Dr. W.J. Canada in 1955,¹³ and since then, only approximately 500 cases have been reported worldwide.^{14,15} CCS is characterized by the presence of multiple hamartomatous polyps throughout the gastrointestinal tract (apart from the esophagus), as well as multiple cutaneous changes, including loss of the hair and nails and hyperpigmentation of the skin. Patients with CCS usually present with diarrhea and protein-losing enteropathy as the dominant gastrointestinal symptoms.^{14,16,17} Patients do not necessarily have the full-blown cutaneous manifestations at presentation, and the severity of the skin changes, which are reversible, may evolve with underlying disease activity.¹⁸

There are no established diagnostic criteria for CCS, and the diagnosis should be considered when clinical characteristics compatible with CCS, such as this patient's gastrointestinal and cutaneous symptoms, are present and when other conditions are ruled out. Then, the diagnosis can be established if multiple hamartomatous polyps are identified in the gastrointestinal tract. This patient has many clinical features of CCS, and an endoscopic examination would most likely reveal multiple polyps that should be resected for histologic examination.

DR. JIMMY C.T. LAI'S DIAGNOSIS

Cronkhite–Canada syndrome.

DIAGNOSTIC TESTING

Dr. Wu: Additional laboratory studies were obtained (Table 2). Tests for autoimmune markers were negative, with the exception of a weakly positive antineutrophil cytoplasmic antibody test with an antiproteinase 3 antibody level of 27.1 RU per milliliter. The anti–tissue transglutaminase IgA level was normal, as were the pituitary hormone levels. The blood level of free triiodothyronine was suggestive of euthyroid sick syndrome. The blood levels of copper, mercury, lead, and selenium were normal. The urine arsenic level was within an acceptable range. The traditional Chinese medicine was analyzed, and its ingredients did not include arsenic or lead. Positron-emission tomography of the skull base to the thighs did not show any hypermetabolic uptake suggestive of cancer.

Dr. Louis H.S. Lau: On the second hospital day, esophagogastroduodenoscopy (EGD) revealed atrophic background gastric mucosa with multiple gastric polyps in clusters. The polyps were erythematous with inflammatory exudates that predominantly affected the antrum and lesser curvature of the stomach (Fig. 2A). There was diffuse villous atrophy in the duodenum (Fig. 2B). Targeted biopsy specimens were obtained from gastric polyps and duodenal mucosa.

A colonoscopy revealed nonspecific colitis, and numerous sessile polyps in clusters were observed throughout the entire colon (Fig. 2C). On white-light examination, the colonic polyps were hyperemic with a strawberry-like appearance (Fig. 2D). With narrow-band imaging, the pit patterns of the polyps were regular and stellar (Kudo classification type II) (Fig. 2E), a finding that was suggestive of inflammatory and non-neoplastic polyps. Snare polypectomy was performed to remove the largest polyps for histopathological examination.

Dr. Howard H.W. Leung: Histologic examination of the gastric, small intestinal, and colorectal polyps revealed hamartomatous features. The mucosal glands showed hyperplastic changes with irregular branching, cystic dilatation, and serration. The stroma appeared to be edematous. The cystic dilated glands and stroma consisted

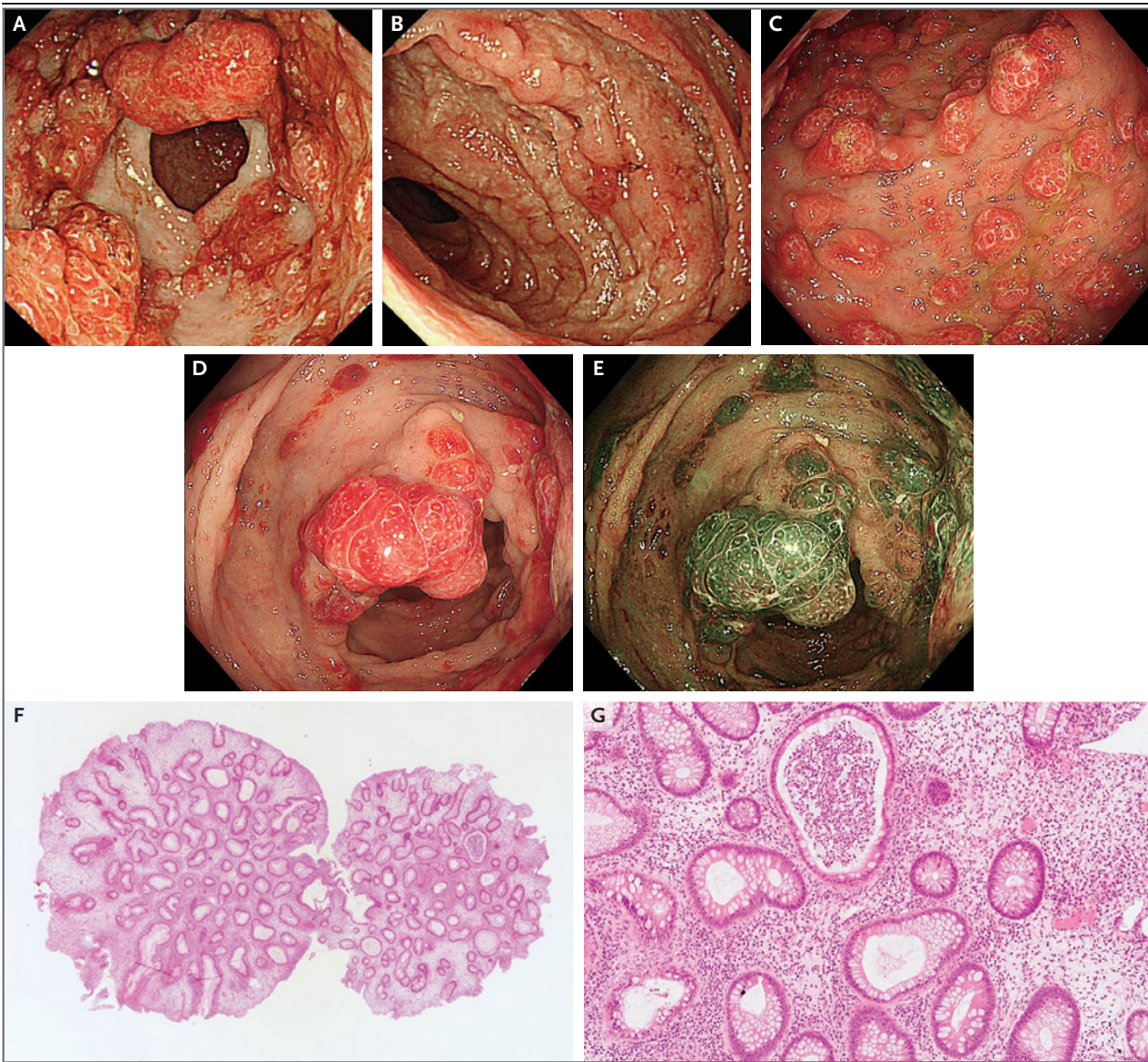
Table 2. Additional Laboratory Data.*

Variable	Reference Range	On Admission
Blood		
Antinuclear antibodies	Negative	Negative
Antineutrophil cytoplasmic antibodies	Negative	Weakly positive
Antiproteinase 3 antibodies (RU/ml)	<20.0	27.1
Anti–double-stranded DNA antibodies (IU/ml)	<100	33
C3 (g/liter)	0.90–1.80	1.19
C4 (g/liter)	0.10–0.40	0.26
Anti–tissue transglutaminase IgA (units)	<20	<4
Anti– <i>Saccharomyces cerevisiae</i> antibodies	Negative	Negative
Free thyroxine (ng/dl)	12.0–22.0	8.8
Free triiodothyronine (pmol/liter)	3.10–6.80	2.76
Thyrotropin (mIU/liter)	0.27–4.20	1.39
Prolactin (mIU/liter)	<496	225
Luteinizing hormone, ovulation phase (IU/liter)	14.0–95.6	18.8
Follicle-stimulating hormone, ovulation phase (IU/liter)	4.7–21.5	7.3
Estradiol, ovulation phase (pmol/liter)	177–1153	956
Corticotropin (pmol/liter)	<10.2	5.8
Copper (μmol/liter)	11.0–25.0	11.5
Selenium (μmol/liter)	0.80–2.00	0.83
Lead (μg/dl)	<10.00	1.91
Mercury (μmol/liter)	<77	4
Urine		
24-Hr mercury (nmol/day)	<50.0	3.2
24-Hr arsenic (nmol/day)	<670.0	87.6
Arsenic-to-creatinine ratio (nmol/mmol)	<68.0	4.8

* To convert the values for free thyroxine to picomoles per liter, multiply by 12.87. RU denotes relative units.

of moderate inflammatory infiltrate, including lymphocytes, eosinophils, plasma cells, and neutrophils (Fig. 2F and 2G). The biopsy specimens of intervening mucosa between the polyps also showed edematous and inflamed lamina propria with mild glandular irregularities. No dysplasia or cancer was noted.

The histologic features were compatible with CCS. The presence of intervening mucosa with morphologic features that were similar to those of the polyps helped differentiate this disease from other non-neoplastic polyposis diseases.



PATHOLOGICAL DIAGNOSIS

Cronkhite–Canada syndrome.

DISCUSSION OF MANAGEMENT

Dr. Lai: CCS is a rare, nonhereditary hamartomatous polyposis syndrome that involves cutaneous changes. Despite being a rare disease, CCS is considered as a possible diagnosis in patients with protein-losing enteropathy and unexplained cutaneous changes because the disease and its management have implications for long-term morbidity and mortality. Even though the patho-

genesis of CCS is not known, most reported treatments involve the use of immunosuppression. Data from prospective trials investigating the treatment of CCS are lacking, given the rarity of this disease; choices of treatment are mainly guided by case reports and case series.

TREATMENT OF *HELICOBACTER PYLORI* INFECTION

The general approach for the treatment of patients with CCS is shown in Table 3. The initial approach to the management of CCS includes treating *H. pylori* infection in patients who have a positive test; this strategy has been shown to result in regression of polyposis in the gastroin-

Figure 2 (facing page). Images Obtained during EGD and Colonoscopy and Resection Specimens.

Images from esophagogastroduodenoscopy (EGD) show multiple inflammatory gastric polyps in the prepyloric antrum (Panel A) and diffuse villous atrophy in the second part of the duodenum (Panel B). Images from colonoscopy show numerous sessile polyps in clusters throughout the entire colon (Panel C). On white-light examination, large colonic polyps that are hyperemic with a strawberry-like appearance are observed (Panel D); with narrow-band imaging, the pit patterns of these polyps are regular and papillary, consistent with Kudo classification type II (Panel E). On hematoxylin and eosin staining, a resection specimen from a polyp from the descending colon shows hamartomatous features, including cystic dilatation of crypts with inflammatory infiltrate in edematous stroma at low magnification (Panel F) and higher magnification (Panel G).

testinal tract and a reduction in skin changes.¹⁹ It is important that patients with gastroesophageal reflux disease also be treated with histamine type 2 receptor antagonists or proton-pump inhibitors.²⁰ This patient's gastric antrum–biopsy specimen showed the presence of *H. pylori*, and she received clarithromycin-based triple therapy (clarithromycin, amoxicillin, and pantoprazole) for 2 weeks. Examination of gastric antrum–biopsy specimens obtained during follow-up EGD confirmed the eradication of the organism.

IMMUNOSUPPRESSION

Immunosuppression is the mainstay of treatment of CCS. The most commonly reported immunosuppressive treatment is systemic glucocorticoids; in case series and retrospective cohort studies, more than half of patients had a treatment response.^{21,22} However, there is no agreement regarding the choice, route, dose, or duration of glucocorticoid therapy. The addition of mesalamine, a 5-aminosalicylate, to systemic glucocorticoid therapy has also been reported.²³ It is notable that there is no agreement regarding the need for glucocorticoid-sparing agents as maintenance therapy, and the incidence of symptom recurrence in patients with an initial glucocorticoid response is not well-described in the literature. This patient received intravenous hydrocortisone, followed by a 2-month tapering course of oral prednisolone. Unfortunately, her clinical condition did not improve.

In patients with glucocorticoid-refractory disease who remain symptomatic with persistent

Table 3. Management of the Cronkhite–Canada Syndrome.**Medical therapy**

- Treatment of *Helicobacter pylori* infection in all patients with a positive test
- Treatment of gastroesophageal reflux disease (histamine type 2 receptor antagonists, proton-pump inhibitors)
- Immunosuppressive treatments on the basis of response
 - Initial trial of systemic glucocorticoids with or without mesalamine
 - If disease is refractory to systemic glucocorticoids, consider azathioprine or cyclosporin A
 - If disease is refractory to azathioprine or cyclosporin A, consider infliximab

Surgical therapy in selected cases**Prevention and treatment of complications**

- Treatment of infection
- Prevention of thrombosis

Nutritional support

- Enteral feeding preferred
- If enteral feeding causes unacceptable adverse effects or if nutritional deficiencies remain despite enteral nutritional support, consider parenteral feeding

skin changes and no regression of gastrointestinal polyposis, the use of nonsteroidal immunosuppressants such as azathioprine and cyclosporin A can be considered.^{24,25} In patients with disease that is refractory to all these treatments, infliximab may be a possible treatment option; a few case reports indicated that treatment with this anti-tumor necrosis factor α monoclonal antibody showed encouraging results, including the alleviation of symptoms and polyp regression.^{26,27} In this patient, after no response to systemic glucocorticoid therapy was seen over the course of 3 months, treatment with cyclosporin A was initiated. When there was no response to cyclosporin A after 1 year, treatment with infliximab was started.

SURGICAL THERAPY

A few case reports of CCS have shown that, after the failure of immunosuppressive therapy with systemic glucocorticoids, colectomy (subtotal colectomy or hemicolectomy) resulted in abatement of diarrhea, gastrointestinal polyposis, and protein-losing enteropathy.^{28,29,30} However, with the availability of medical therapy for CCS, which involves not only the colon but also the stomach and small bowel, the surgical option is reserved primarily for patients with refractory CCS or for those with neoplastic complications.

NUTRITIONAL SUPPORT

In addition to the described specific treatments for coexisting conditions and the administration of immunosuppressive therapy, general supportive measures are of paramount importance in caring for patients with CCS. It is vital that patients receive maximal nutritional support and electrolyte replacement. If enteral feeding causes severe gastrointestinal symptoms to occur, or if there are refractory nutritional deficiencies despite the receipt of enteral nutritional support, then parenteral nutrition is warranted. This patient received intensive oral nutritional and electrolyte supplementation; however, her nutritional status remained poor with continued electrolyte imbalances. As a result, total parenteral nutrition was started.

PREVENTION OF COMPLICATIONS

Patients with CCS are susceptible to infections because of weakened immunity associated with a loss of immunoglobulins caused by protein-losing enteropathy and because of the use of immunosuppressants as the mainstay of treatment. This patient had sepsis and recurrent *C. difficile* infections that were treated with antibiotic agents and fecal microbiota transplantation. Fecal microbiota transplantation is a well-established treatment for patients with recurrent or refractory *C. difficile* infection, but its effect in patients with underlying CCS is unclear.³¹ This patient received fecal microbiota transplantation twice (through the upper and lower gastrointestinal tract), and the *C. difficile* infection resolved.

The patient also had deep venous thrombosis and pulmonary embolism, for which she received anticoagulant therapy. The hypercoagulable state associated with CCS is due to a loss of intrinsic anticoagulants, such as protein C and protein S, along with elevated fibrinogen and factor VIII levels.³² Hence, it is important to consider prophylactic anticoagulation in patients with CCS.

MONITORING OF DISEASE ACTIVITY

Even though there are no standardized guidelines for treatment or monitoring in patients with CCS, regular review of the patient's clinical, biochemical, and endoscopic status is warranted. Reports have suggested that the evolution of skin changes reflects underlying disease activity.¹⁸ For example, patients with controlled disease activity may have an improvement in

skin color and regrowth of hair and nails. Resolution of abdominal pain and diarrhea, as well as weight gain, also reflect improved disease control. With better disease control, laboratory testing may show improved blood levels of albumin and electrolytes. If there is a decrease in disease activity, the intensity of supportive treatments including nutritional support and electrolyte replacement can be reduced. Finally, regular assessment with the use of upper and lower endoscopy can be considered to assess the polyposis directly. Surveillance endoscopy can be used to assess the number of polyps, which can reflect disease control, as well as to detect the development of neoplastic complications, which are inherent in polyposis syndromes.^{30,33} Unfortunately, the overall prognosis of CCS is poor, with a reported 5-year mortality of 55%, mostly due to complications such as sepsis and gastrointestinal bleeding.¹⁵

FOLLOW-UP

Dr. Wu: After more than a year of treatment with prednisolone and then cyclosporin, which resulted in a suboptimal clinical response, treatment with infliximab was started. After the patient began treatment with infliximab, she noticed an improvement in the cutaneous features. The skin pigmentation was lighter, and there was regrowth of hair that was much thicker and less brittle. Follow-up EGD and colonoscopy revealed partial regression of the multiple gastrointestinal polyps.

However, the patient had persistent nonbloody diarrhea up to eight times a day and swelling of the legs. She was dependent on total parenteral nutrition. The blood levels of calcium and vitamin D remained low despite regular and intensive oral and intravenous calcium and phosphate supplementation, as well as oral and intramuscular vitamin D supplementation. Severe osteomalacia developed, and she sustained nontraumatic fractures, including multiple rib fractures that had led to flail chest and hypoventilation syndrome. She also had scapular and clavicular fractures and multiple nondisplaced pelvic fractures. These fractures were deemed to be inoperable by orthopedic and cardiothoracic surgical teams.

The patient had recurrent pulmonary embolism and recurrent pneumonia. Thirty months

after the diagnosis of CCS, the patient had respiratory failure and died.

FINAL DIAGNOSIS

Cronkhite–Canada syndrome.

This case was presented as a special feature of the 100-year anniversary of the Case Records of the Massachusetts General Hospital.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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